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History of Astronomy: The 19th Century

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Introduction

Imagine a girl born in the year 1800, on the night her town's church bells rang in the new century. As an infant, she slept under a sky that astronomers still measured mostly with brass instruments and brass instruments alone: naked eyes, pendulum clocks, and star charts drawn by hand. By the time she was an old woman in 1900, astronomy had photographed the Moon, weighed the Sun, measured the distance to the stars, discovered a planet with mathematics before anyone saw it, and learned what the stars are made of from the light they sent across the void. No other century transformed our picture of the sky so completely.

The nineteenth century is when astronomy stopped being only the science of WHERE things are and became, in addition, the science of WHAT things are. For more than two thousand years, astronomers had charted positions: the wandering planets, the wheeling stars, the slow drift of the equinoxes. That work reached its spectacular climax in the 1800s, when the mathematics of gravity grew so precise that it could point a telescope at an empty patch of sky and say: look here, and you will find a new world. It did, and the world was Neptune.

But the same century opened a second door entirely. When chemists learned to read the rainbow of sunlight, and inventors found ways to paint with light itself on polished metal and glass, astronomers gained tools that turned starlight into information. A dark line in a spectrum could reveal the iron in a star's atmosphere. A photograph could hold a thousand stars in a single image, patiently, without fatigue. The questions changed overnight. What are the stars made of? How far away are they? How fast do they move? Are the nebulae clouds of gas or cities of suns? None of these questions had been answerable before 1800. By 1900, all of them had answers, or at least the beginnings of answers.

This book tells the story of that transformation through its people and its instruments. You will meet the unfailingly polite Englishman who calculated a new planet and was too shy to publish loudly enough, and the bold Frenchman who refused to be ignored. The orphaned glassworker's apprentice who mapped hundreds of dark lines in sunlight and died young, leaving optics that no one could equal for a generation. The Irish lord who built a telescope the size of a warship cannon in his own front garden and first saw that some nebulae are spiral, like vast wheels of stars. The widow's daughter hired to do arithmetic at an observatory, who ended up classifying the stars of the heavens. And the rich and eccentric man who wanted the largest telescope on Earth, and who lies buried beneath it still.

The story is told here for curious readers of every age. No mathematics is needed beyond what a garden and a lamppost can teach, and every technical idea is explained when it appears. Chapter by chapter, we will follow the century's great arc: its triumphant opening, when the heavens surrendered their positions to calculation; its middle years, when new instruments cracked open the light of the sky; and its closing decades, when astronomy handed the twentieth century a ready-made toolkit of photography, spectroscopy, and giant telescopes, pointed at questions no one had even known how to ask in 1800.

One warning before we begin. The nineteenth century was also a century of confident mistakes, and we will meet them honestly: the planet Vulcan that never existed, the canals of Mars that were an illusion of the eye, and a famous age-of-the-Earth calculation that made the Sun far too young. Science's power to correct itself is part of this story too. The century that discovered Neptune with a pen also, slowly and painfully, learned to distrust the pen when the sky disagreed. That double lesson, confidence checked by evidence, is the true inheritance of the nineteenth century, and it is the one this book most wants to hand to you.

Chapter 1: The Discovery of The 19th Century

Every century has a moment that comes to define it, and for the astronomers of the 1800s that moment arrived on the night of September 23, 1846, in Berlin. At the New Observatory, a young astronomer named Johann Galle opened a letter that had arrived that very day. It came from France, from a mathematician named Urbain Le Verrier, and its message was extraordinary: the planet Uranus, Galle's own generation's proudest discovery, was being tugged off its predicted course by something unseen, and Le Verrier had calculated exactly where that something would appear in the sky. Point your telescope, the letter said, within a degree of a particular patch of stars in the constellation Aquarius, and you will find a new planet.

Galle and his student assistant, Heinrich d'Arrest, did exactly that. Working the same night, they compared the stars in the telescope's view against a fine new chart of that region, and within the hour d'Arrest spoke the quiet, historic sentence: "That star is not on the map." It was not a star. It was the planet we now call Neptune, hanging

almost exactly where Le Verrier's mathematics said it must be. A new world had been discovered not by sweeping the sky with the eye, but by thinking about it. It was the first planet ever found with the point of a pen.

To understand what a thunderclap this was, we must back up to 1781. In that year, the musician turned astronomer William Herschel, sweeping the sky from his garden in Bath, England, had found Uranus, the first new planet of recorded history. After the initial excitement, astronomers began the patient work of calculating its orbit, and at first the mathematics behaved. Year after year, Uranus tracked the path drawn for it. But as decades passed and observations piled up, a stubborn fact emerged: Uranus was drifting. Not much, only tiny amounts, but far more than any honest measurement error could explain. By the 1830s and 1840s, the mismatch between where Uranus was and where Newton's law of gravity said it should be had grown into a genuine scandal. Some astronomers even suggested, quietly and uncomfortably, that perhaps Newton's law of gravity failed at such enormous distances.

The correct answer had been suggested as early as the 1820s: perhaps an unseen planet beyond Uranus was pulling on it. But suggesting is not solving. To find the culprit, you must know where to point the telescope, and the sky is enormous. The needed calculation was ferociously difficult: you had to work backward from Uranus's misbehavior and deduce, from its effects alone, both the distance and the position of a planet no one had ever seen. Two young mathematicians, unknown to each other, took up the challenge.

In England, the task obsessed a brilliant, shy Cambridge graduate named John Couch Adams. Working alone across several years in the early 1840s, and continuing his calculations while tutoring in Cornwall, Adams gradually built a mathematical portrait of the unseen planet, including where it should be found. In the autumn of 1845 he presented his results to the Astronomer Royal, George Airy, in London. What followed became one of the saddest and most argued-about episodes in the history of science. Airy was polite but did not treat the prediction as urgent, no prompt and serious telescope search was begun in Britain, and Adams's mathematics was left sitting in official files while the sky waited.

In France, Urbain Le Verrier, a rising star of celestial mechanics, attacked the same problem with fierce confidence and, unlike Adams, a genius for making the world pay attention. In June 1846 he published his predicted position for the unseen planet, and when the Paris Observatory did not immediately act on it, he sent his letter to Berlin. Galle found Neptune within a degree of Le Verrier's calculated spot, a single night after the letter arrived. In England, the realization that Adams had arrived at nearly the same answer months earlier triggered shock, soul-searching, and a long priority dispute that even strained relations between the two nations' astronomers. Today, historians and astronomers give both men their due: Adams and Le Verrier each calculated the planet's position independently, and the discovery belongs to the mathematics itself, verified at the telescope by Galle.

The discovery of Neptune transformed what astronomers believed mathematics could do, and the mood of the century followed it. If a planet could be conjured out of equations, what else might calculation summon from the dark? The answer came quickly and, this time, wrongly. Le Verrier, now the world's most celebrated celestial mechanic, noticed that the planet Mercury also misbehaved, wobbling in its orbit by a small amount his tables could not explain. With Neptune's triumph fresh in everyone's memory, he announced a solution: another unseen planet, smaller and closer to the Sun, circling inside Mercury's orbit. Its name was already chosen, Vulcan, for the god of the forge, fitting for a planet forever hugging the fires of the Sun. Astronomers hunted for it for decades, during eclipses, in daylight, and in the memoirs of a country doctor named Lescarbault, who claimed to have seen it pass across the Sun in 1859. We will return to Vulcan in later chapters, because the story of how it refused to exist, and of what finally explained Mercury's wobble sixty years later, is one of the century's greatest lessons in scientific patience.

The 1846 discovery also crowned the career of the eighteenth century's own mathematics. Newton had promised that gravity governed the heavens, and Neptune was the promise kept at full interest: a prediction so precise that a telescope could almost have found the planet with its lens cap on. When the British astronomer John Herschel (son of Uranus's discoverer) described the discovery, he said the new planet had been found "with the point of the pen," and the phrase became the century's proudest boast. Yet even as the champagne was poured in Berlin, the seeds of a different astronomy were sprouting elsewhere: in a Munich workshop where a young optician was polishing lenses of unheard-of quality, and in a laboratory where a chemist was about to notice that a candle flame held strange signatures in its colors. The story of the pen had reached its summit. The story of light was about to begin.